Bayesian Logistic Regression and Hierarchical Modeling for Alzheimer's Disease Prediction

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Abstract

Alzheimer's disease is a major cause of dementia worldwide, and early prediction is vital for timely intervention. Using demographic, lifestyle, medical history, and cognitive assessment data from $N = 2149$ patients, we developed and compared two Bayesian models for predicting Alzheimer's disease diagnosis; baseline logistic regression and a hierarchical extension accounting for ethnicity-specific variation. Bayesian inference enables probabilistic prediction and uncertainty quantification while incorporating weakly informative priors to regularize estimation. The strongest Risk predictors were Memory Complaints and Behavioral Problems, whereas Mini-Mental State Examination (MMSE), Activities of Daily Living (ADL), and Functional Assessment scores showed protective effects. Leave-One-Out cross-validation (LOO-CV) revealed that although the Hierarchical model captured moderate variation across four ethnicity groups ($\sigma_\alpha = 0.41$), it did not improve predictive performance over the baseline model. These findings suggest that the simpler model achieves equivalent predictive accuracy with greater computational efficiency for clinical applications, while the hierarchical approach remains valuable for population-level research and subgroup analysis.

1. Introduction

Alzheimer's disease (AD) is a progressive neurodegenerative disorder and one of the leading causes of disability and mortality among older adults (1). Early identification of individuals at elevated risk can improve clinical decision-making and enable preventive strategies that may delay cognitive decline. Predictive modeling has therefore become a central tool in AD research.

Traditional logistic regression provides point estimates but fails to fully characterize uncertainty in predicted risks, which is critical in a clinical setting. Bayesian methods overcome this limitation by delivering full posterior distributions and naturally incorporating prior knowledge (3). Bayesian inference also mitigates overfitting through prior regularization, which is beneficial in high-dimensional health datasets.

We examine probabilistic prediction of AD diagnosis based on demographic, lifestyle, historical medical conditions, and cognitive assessment variables. We fit a Bayesian logistic regression model and extend it to a hierarchical structure that accounts for baseline differences across four patient ethnicity groups: Caucasian, African American, Asian, and Other. Hierarchical models enable partial pooling across groups and are well-suited for heterogeneous populations (7; 2).

We evaluate and compare the models using posterior predictive checks, Leave-One-Out cross-validation (LOO-CV), and Widely Applicable Information Criterion (WAIC) to assess predictive performance. Our analysis addresses the following questions:

- Which clinical and lifestyle factors are most strongly associated with elevated AD risk?
- Does modeling ethnicity as a group-level random effect improve predictive accuracy?
- What are the methodological trade-offs between hierarchical and non-hierarchical Bayesian models in this context?

Our results contribute to the development of robust probabilistic tools for clinical risk stratification and highlight the importance of cognitive measures in AD diagnosis prediction.

2. Data

The dataset consists of $N = 2149$ older adults aged 60–90 years, including detailed demographic, lifestyle, medical history, and cognitive assessment information. The binary outcome variable indicates whether an individual is diagnosed with Alzheimer's disease (AD).

The following groups of predictors were included:

- **Demographic**: Age, gender, education level, and ethnicity.

- **Lifestyle**: Smoking, physical activity, sleep quality, BMI.
- **Medical history**: Cardiovascular disease, hypertension, diabetes, depression, family history of AD, head injury.
- **Cognitive and functional measures**: MMSE, Functional Assessment, Activities of Daily Living (ADL), Memory Complaints, Behavioral Problems.

Ethnicity was coded into four groups for hierarchical modeling:

0: Caucasian 1: African American 2: Asian 3: Other

Continuous predictors (e.g., MMSE, ADL, BMI) were standardized to mean zero and unit variance following Stan recommendations to improve sampler geometry and interpretation.

3. Models and Methods

3.1. Baseline Bayesian Logistic Regression

Let $y_i \in \{0, 1\}$ represent AD diagnosis for patient $i = 1, \ldots, N$. Covariates are divided into binary predictors: x_{ij}, $j = 1, \ldots, J$ and standardized continuous predictors: z_{ik}, $k = 1, \ldots, K$

$$y_i \sim \text{Bernoulli}(p_i), \qquad \text{logit}(p_i) = \alpha + \sum_{j=1}^{J} \beta_j x_{ij} + \sum_{k=1}^{K} \gamma_k z_{ik}. \tag{1}$$

Prior specification. Weakly informative priors provide regularized effect estimation:

$$\alpha \sim \mathcal{N}(\text{logit}(0.354), 2), \tag{2}$$
$$\beta_j \sim \mathcal{N}(0, 1.5), \tag{3}$$
$$\gamma_k \sim \mathcal{N}(0, 1.5). \tag{4}$$

where 0.354 is the empirical positive class prevalence, centering the intercept on a realistic prior belief.

This model yields probabilistic predictions that directly quantify uncertainty in AD risk.

3.2. Hierarchical Logistic Regression by Ethnicity

To allow baseline risk to vary across ethnicity groups, we extend the linear predictor with a random intercept:

$$\text{logit}(p_i) = \mu_\alpha + \alpha_{g[i]} + \sum_{j=1}^{J} \beta_j x_{ij} + \sum_{k=1}^{K} \gamma_k z_{ik}, \tag{5}$$

where group membership $g[i] \in \{1, \ldots, 4\}$ indexes ethnicity.

$$\alpha_g \sim \mathcal{N}(0, \sigma_\alpha), \tag{6}$$
$$\mu_\alpha \sim \mathcal{N}(0, 2), \tag{7}$$
$$\sigma_\alpha \sim \text{Half-Normal}(1). \tag{8}$$

This induces partial pooling, meaning estimates for smaller ethnic groups borrow strength from the population while retaining group-specific differences (2).

3.3. Posterior Inference and Model Comparison

Posterior sampling was implemented via a Markov Chain Monte Carlo (MCMC) in Stan using the `rstan` interface. The models reliability were assessed through: $\hat{R} \approx 1.00$ for all parameters indicating convergence, Traceplots confirming good chain mixing and Posterior predictive checks based on replicated y_{rep}.

Predictive performance was compared using: Leave-One-Out Cross-Validation (LOO-CV) and Widely Applicable Information Criterion (WAIC). This was computed using the `loo` package in R.

3.4. Interpretation of Effects

Posterior logistic regression coefficients are presented on the log-odds scale. We interpret them using:

$$\Delta p \approx \text{logit}^{-1}(\alpha + \beta) - \text{logit}^{-1}(\alpha) \tag{9}$$

to clarify clinical impact by converting selected effects into probability differences in the Results section.

4. Results

4.1. Model Convergence and Reliability

Both Bayesian models exhibited excellent convergence, with all parameters achieving $\hat{R} \approx 1.00$. Traceplots demonstrated good chain mixing (see Appendix). Posterior predictive checks confirmed that both models accurately reproduced the observed outcome distribution, indicating appropriate calibration.

4.2. Key Predictors of Alzheimer's Disease Risk

Table 1 summarizes the posterior distributions for the coefficients with the largest magnitude of effect. Memory Complaints and Behavioral Problems were the strongest risk factors, substantially increasing the log-odds of AD diagnosis. Higher MMSE, ADL, and Functional Assessment scores were associated with lower risk.

We translated selected log-odds effects into clinically relevant probability changes. Memory Complaints: an esti-

mated increase of approximately $+38$ percentage points in probability of AD (holding other predictors constant) Behavioral Problems: approximately $+36$ percentage points increase. A one standard deviation increase in MMSE corresponds to about -5 percentage points in AD probability

These interpretations demonstrate the practical utility of Probabilistic modeling in clinical screening.

Table 1. Posterior means and 95% credible intervals for selected Bayesian logistic regression coefficients. Negative values indicate protective effects.

Predictor	Mean	SD	2.5%	97.5%
Memory Complaints	2.60	0.17	2.27	2.94
Behavioral Problems	2.49	0.18	2.14	2.85
Functional Assessment	-0.45	0.03	-0.50	-0.40
ADL	-0.42	0.03	-0.47	-0.37
MMSE	-0.11	0.01	-0.12	-0.09
Head Injury	-0.31	0.23	-0.76	0.13
Smoking	-0.21	0.14	-0.48	0.06

4.3. Classification Performance and Predictive Utility

For clinical interpretability, we report classification accuracy, sensitivity, specificity, and precision using a 0.5 threshold (Table 3). The baseline model achieved an accuracy of 0.845 with moderately high specificity.

Posterior predictive calibration curves (Appendix) confirmed good concordance between predicted probabilities and observed outcomes.

We additionally evaluated discrimination performance using the Area Under the ROC Curve (AUC). Both models achieve an estimated AUC of approximately 0.86, indicating a strong ability to separate AD and non-AD individuals.

Table 2. Confusion matrix for baseline model predictions (threshold $= 0.50$).

	Pred. 0	Pred. 1
Obs. 0	1315	259
Obs. 1	74	501

4.4. Hierarchical Model Results

The ethnicity-specific intercept variation was captured by the group-level standard deviation:

$$\sigma_\alpha = 0.41 \;\; (90\% \text{ CrI: } 0.07,\ 0.91),$$

indicating moderate but uncertain heterogeneity in baseline AD risk across the four groups.

The fixed effects remained consistent in magnitude and direction (Appendix Table A1), demonstrating robustness of core clinical predictors.

4.5. Model Comparison Using LOO-CV

Baseline model: *elpd* = 0.0 (reference) and Hierarchical model: $\Delta elpd = -0.1 \pm 1.3$

The difference in predictive performance is negligible, confirming that the additional complexity, while offering interpretability gains, does not improve patient-level prediction.

The WAIC comparison yielded similar conclusions.

4.6. Prior Sensitivity Analysis

We examined posterior stability by moderately tightening and loosening prior scales on regression coefficients. Parameter rank-ordering and posterior uncertainty changed minimally, indicating that the data are highly informative relative to the priors.

Table 3. Confusion matrix for Hierarchical model predictions.

	Pred. 0	Pred. 1
Obs. 0	1312	258
Obs. 1	77	502

4.7. Summary of Findings

- Cognitive impairment measures (Memory Complaints, Behavioral Problems) are the strongest risk indicators.
- Functional and MMSE scores provide protective information.
- Hierarchical modeling reveals meaningful, though uncertain, differences in ethnic baseline risk.
- Predictive accuracy is strong (AUC ≈ 0.86), with limited benefit from hierarchical structure for pure prediction tasks.

5. Conclusions and Discussion

This study developed and compared two Bayesian models to estimate Alzheimer's disease (AD) risk using demographic, lifestyle, medical history, and cognitive assessment variables. Our findings reinforce the strong role of cognitive measures, particularly Memory Complaints and Behavioral Problems are reliable early indicators of AD, in line with prior clinical research (5). Functional and MMSE scores consistently showed protective effects, emphasizing the importance of ongoing cognitive monitoring in older adults.

The hierarchical modeling framework allowed for estimation of ethnicity-specific baseline risks through partial pooling. The moderate estimate of between-ethnicity variation highlights population differences that may reflect real disparities in disease prevalence, structural risk factors, or diagnostic patterns. However, the uncertainty in σ_α and minimal change in predictive performance indicate that baseline heterogeneity alone does not materially enhance clinical risk prediction compared to the simpler logistic model.

Methodologically, this project demonstrates key advantages of Bayesian inference in healthcare prediction contexts: (i) uncertainty quantification through full posterior distributions, (ii) regularization via weakly informative priors, and (iii) flexible model structure enabling subgroup analysis and information sharing across populations (3; 4).

Despite strong discrimination performance (AUC ≈ 0.86), specificity exceeded sensitivity, suggesting that the model is more accurate for identifying non-AD individuals than detecting early disease. Alternative probability thresholds or decision-analytic metrics may better optimize screening utility in clinical settings. Future extensions could examine whether specific predictors exhibit heterogeneous effects across ethnic groups by including interaction terms or varying slopes, or by adopting deeper hierarchical structures.

Further, future work may incorporate longitudinal trajectories of cognitive decline, or imaging and biomarker data when available, which are known to enhance predictive accuracy (6). Incorporating external validation datasets would also support generalizability. Finally, assessing model decision consequences using clinical loss functions (e.g., weighting false negatives more heavily) could improve practical deployment.

In conclusion, when predictive accuracy and computational efficiency are priorities, the baseline Bayesian logistic regression model offers the most appropriate balance. Hierarchical modeling remains valuable for understanding variation across groups and exploring potential health inequities but does not substantially improve prediction in this dataset. Our Bayesian framework provides a rigorous foundation for more refined and personalized AD risk assessment tools in future research.

Acknowledgements

We hereby grant our consent for the utilization of this project report as a reference material within the context of future editions of the course.

References

[1] Alzheimer's Association. 2024 alzheimer's disease facts and figures. *Alzheimer's & Dementia*, 20(1):1–100, 2024.

[2] Andrew Gelman. Multilevel (hierarchical) modeling: What it can and cannot do. *Technometrics*, 48(3):432–435, 2006.

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[5] Clifford R. Jack Jr et al. Nia-aa research framework: Toward a biological definition of alzheimer's disease. *Alzheimer's & Dementia*, 14(4):535–562, 2018.

[6] Gill Livingston et al. Dementia prevention, intervention, and care: 2020 report of the lancet commission. *The Lancet*, 396(10248):413–446, 2020.

[7] Stan Development Team. Prior choice recommendations. `https://mc-stan.org`, 2023.

Appendix A: Stan Model Code Logistic Regression and Hierarchical Model

```
data {
 int<lower=1> N;   // observations
  int<lower=0> J;  // X predictors
  int<lower=0> K;  // Z predictors
  matrix[N, J] X; // matrix for beta_j
  matrix[N, K] Z; // matrix for gamma_k
  int<lower=0,upper=1> y[N];   // outcomes
  real<lower=0,upper=1> p_prior_mean;
  // the p used in alpha

}

parameters {
  real alpha;  // intercept on logit scale
  vector[J] beta; // coefficients for X
  vector[K] gamma; // coefficients for Z
}
transformed parameters {
  vector[N] pi;
 pi = alpha + X * beta + Z * gamma;
 // linear predictor
}

model {
  // Priors
  alpha ~ normal(p_prior_mean, 2);
  beta  ~ normal(0, 1.5);
  gamma ~ normal(0, 1.5);

  // Likelihood
  y ~ bernoulli_logit(pi);

}
generated quantities {
  vector[N] log_lik;
  int y_rep[N];
  for (n in 1:N) {
    log_lik[n] = bernoulli_logit_lpmf(y[n] | pi[n]);
    y_rep[n] = bernoulli_logit_rng(pi[n]);
    // posterior predictive draws
  }
}
```

```
// hierarchical_model.stan
data {
  int<lower=1> N;  // observations
  int<lower=1> J;  // binary predictors
  int<lower=1> K;  // continuous predictors
  int<lower=1> G;  // ethnicity groups
  int<lower=1, upper=G> group[N];
  matrix[N, J] X;  // binary predictors
  matrix[N, K] Z;  // continuous predictors
  int<lower=0, upper=1> y[N];  // outcome
  real<lower=0, upper=1> p_prior_mean;
}

parameters {
  real mu_alpha;   // global mean intercept
  real<lower=0> sigma_alpha;  // SD intercepts
  vector[G] alpha_raw;  // standardized ethnicity
  vector[J] beta;       // binary predictor
  vector[K] gamma;      // continuous predictor
}

transformed parameters {
  vector[G] alpha;
}

model {
  // Hyperpriors
  mu_alpha ~ normal(0, 2);
  sigma_alpha ~ normal(0, 1);

  // Priors
  alpha_raw ~ normal(0, 1);
  beta ~ normal(0, 1.5);
  gamma ~ normal(0, 1.5);

  // Likelihood
  y ~ bernoulli_logit(alpha[group] + X * beta +
  Z * gamma);
}

generated quantities {
  vector[N] log_lik;
  int y_rep[N];
  vector[N] pi;

  for (n in 1:N) {
    pi[n] = alpha[group[n]] + X[n] * beta +
    Z[n] * gamma;
    log_lik[n] = bernoulli_logit_lpmf(y[n] | pi[n])
    y_rep[n] = bernoulli_logit_rng(pi[n]);
  }
}
```

Appendix B: MCMC Diagnostics and Posterior Predictive Checks

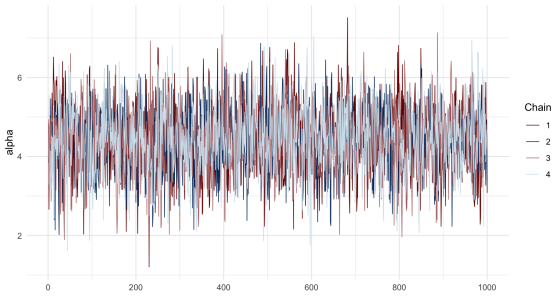


Figure 1. MCMC traceplots for the intercept (α)

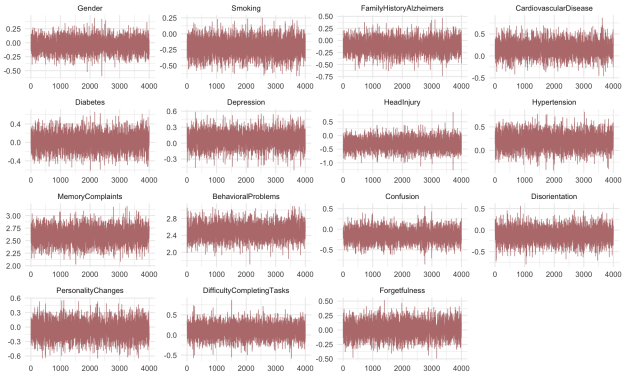


Figure 2. MCMC traceplots for a selection of binary predictor coefficients (β)

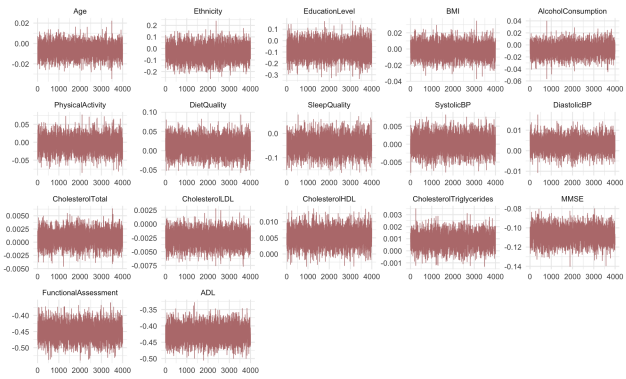


Figure 3. MCMC traceplots for a selection of non-binary predictor coefficients (γ)

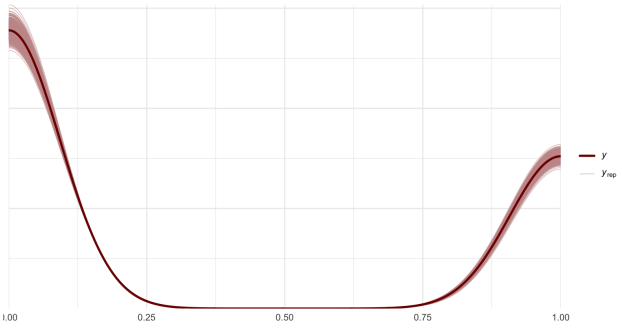
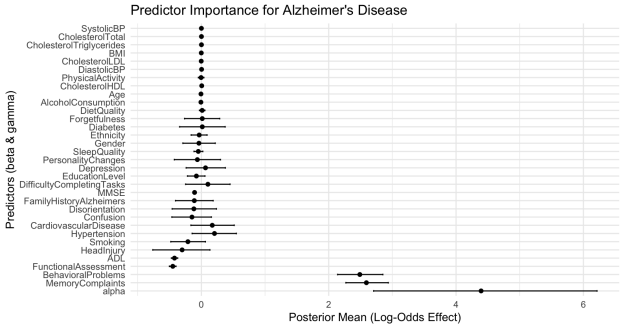


Figure 4. Posterior predictive check: Density overlay of observed outcomes (y) versus posterior predictive replicates (y_rep

Appendix C: Predictor Effects with 95% Credible Intervals



Predictor effects on Alzheimer's disease risk with 95% credible intervals.

Appendix D: Hierarchical Model Diagnostics

6. Hierarchical Model Fixed Effects

Table 4. Posterior summaries for key predictors in the hierarchical model.

Predictor	Mean	SD	90% CrI
Memory Complaints	2.60	0.16	[2.38, 2.81]
Behavioral Problems	2.49	0.18	[2.26, 2.72]
Functional Assessment	-0.45	0.03	[-0.49, -0.42]
ADL	-0.42	0.03	[-0.46, -0.39]
MMSE	-0.11	0.01	[-0.12, -0.10]

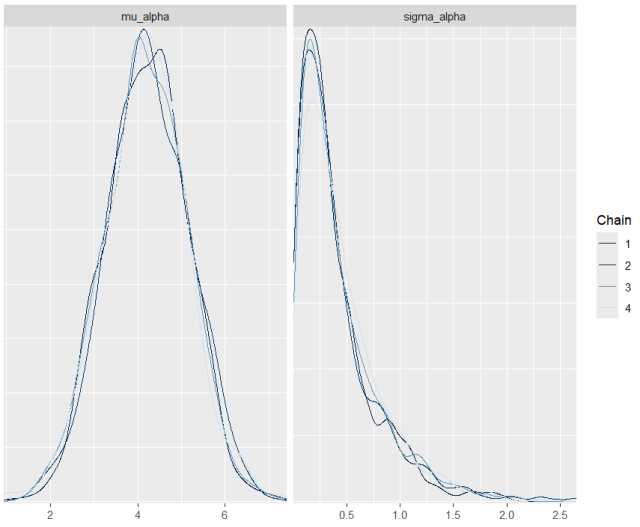


Figure 5. MCMC traceplots for hierarchical model parameters. Left: Global intercept (μ_α) showing excellent chain mixing and convergence. Right: Between-ethnicity standard deviation (σ_α) demonstrating consistent estimates across chains.

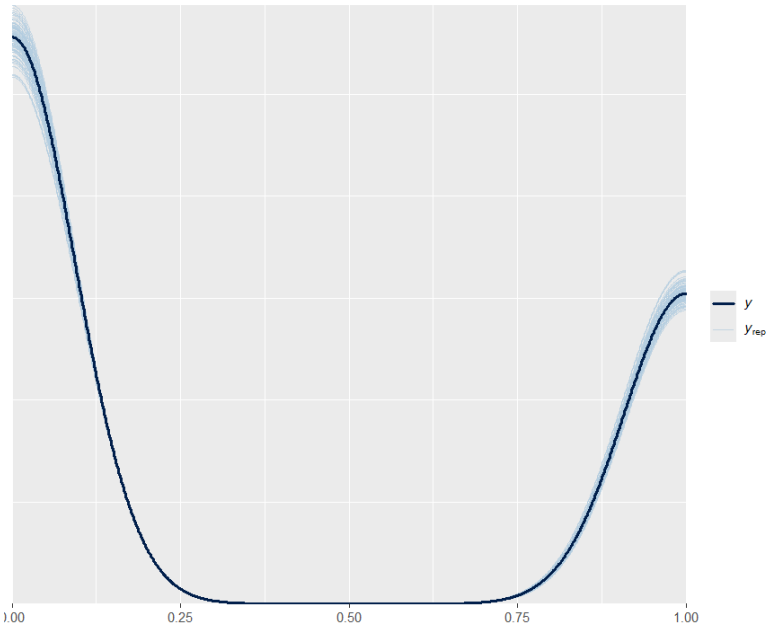


Figure 6. Posterior predictive check for hierarchical model. The density overlay shows close agreement between observed outcomes (y) and posterior predictive replicates (y_rep), indicating good model calibration.